

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and can cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/57/57-007-037-tr.html>)
- Remove the oil pan and gasket. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/57/57-007-025-tr.html>)
- Remove the oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/57/57-007-035-tr.html>)

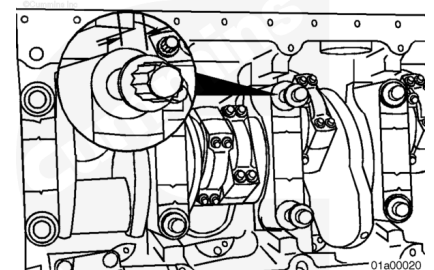
Remove

⚠ CAUTION ⚠

Replace the bearings one pair at a time. Personal injury and damage to the crankshaft can result if the crankshaft falls.

⚠ CAUTION ⚠

The caps must be pulled into position against the block before the crankshaft is rotated. As bearings are replaced, tighten the cap to a minimum of 68 N•m [50 ft-lb] before turning the crankshaft.

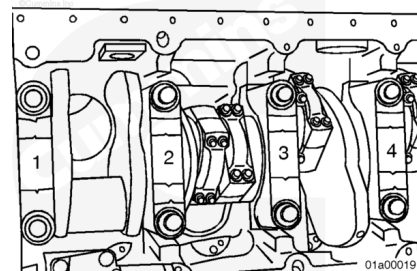


Loosen the main bearing cap capscrews until there is approximately 13 mm [1/2 in] between the capscrew head and the cap. Use two pry bars to loosen the cap. Put the pry bars in the notches at the sides of the caps.

The main bearing caps **must** be marked for position. The last number on each bearing cap identifies the cap location in the block.

The block and main bearing cap identification numbers **must** be identical.

Use a steel stamp to mark any cap that is **not** marked correctly.

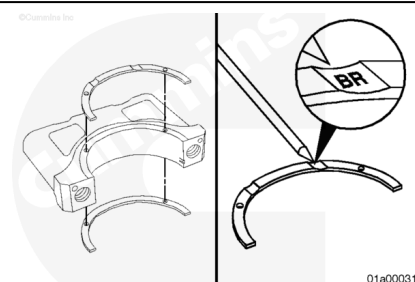


Remove the two lower thrust bearings from the number 7 main bearing cap. Mark each bearing's position in the notched area.

Note : Thrust bearing size is stamped on a crankshaft counterweight adjacent to the thrust bearing location.

Note : Only three bearings are used: One each in the upper and lower rear and one in the lower front.

Note : The three thrust bearings are doweled to the number 7 main cap and saddle.

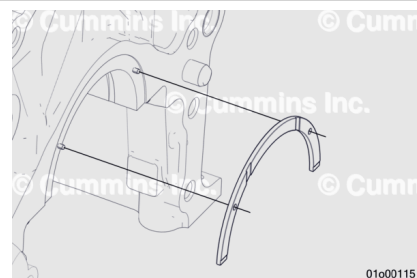


Note : The crankshaft must be removed before the upper thrust bearing can be removed, as it is doweled to the number 7 main saddle at the factory.

Remove the crankshaft. Refer to Procedure 001-016 (Crankshaft) in Section 1.

(/qs3/pubsys2/xml/en/procedures/57/57-001-016-tr.html)

Remove the upper thrust bearing from the block at the number 7 location. Mark for position.



Clean and Inspect for Reuse

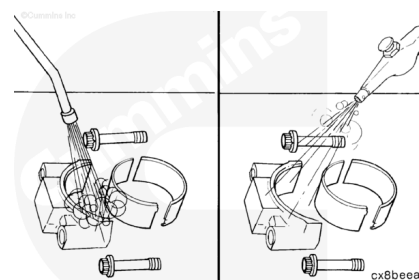
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the

possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

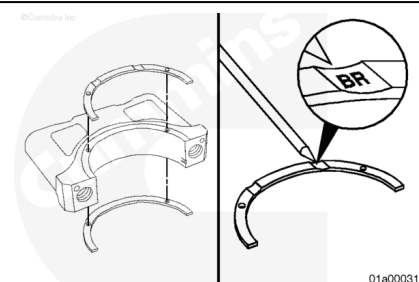


Use solvent to clean the bearings.

Dry with compressed air.

Thrust bearings are available in five sizes: Standard, 0.25 mm, 0.50 mm, 0.75 mm, and 1.00 mm oversize. The upper and lower thrust bearings **must** be the same size. The front and rear can be different sizes.

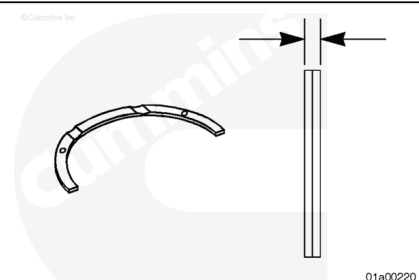
Use the same size as those removed. The crankshaft will be marked on a counterweight adjacent to the thrust location if the thrust flange has been machined for oversize thrust bearings.



Measure the thrust bearing thickness. Measure the areas that shine. The shine indicates where the crankshaft has contacted the bearing.

Thrust Bearing Thickness Standard or Oversize (OS)
(new bearings)

	mm		in
Standard	3.880	MIN	0.1528
	3.930	MAX	0.1547
0.25 mm (OS)	4.005	MIN	0.1577
	4.055	MAX	0.1596
0.50 mm (OS)	4.130	MIN	0.1626
	4.180	MAX	0.1646
0.75 mm (OS)	4.255	MIN	0.1675
	4.305	MAX	0.1695
1.00 mm (OS)	4.380	MIN	0.1724
	4.430	MAX	0.1744

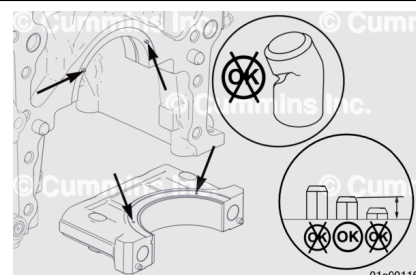


Note : Thrust bearings can be reused with the same components and in the same location from which they were removed, provided they are **not** damaged and the wear does **not** exceed 0.127 mm [0.005 in] less than the minimum thickness of a new bearing. **Example:** A new standard thrust bearing minimum thickness is 3.880 mm [0.1528 in]. Acceptable reuse thickness is 3.753 mm [0.1478 in].

Check the thrust bearing retaining pins in the main bearing cap and block for mechanical damage.

Check thrust bearing retaining pins protrusion on both sides of main bearing cap.

Check thrust bearing retaining pins protrusion on the block.



Retaining Pin Protrusion

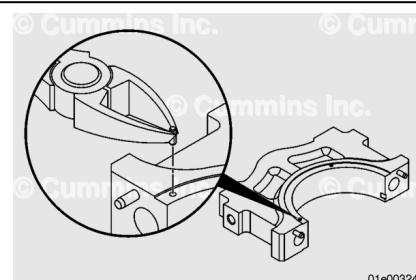
mm		in
2.7	MIN	0.106
3.4	MAX	0.133

If the thrust bearing retaining pins are damaged or have incorrect protrusion, they **must** be replaced.

Replace

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Note : If one or more pins are damaged, replace all four pins.

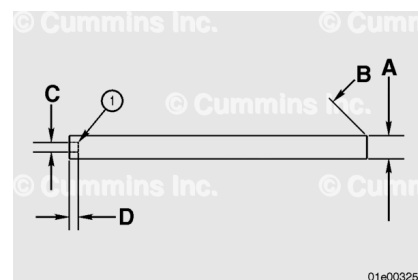
Use proper tooling to grip the damaged pin. Place a piece of thin wood, or equivalent, between the gripping tool and main bearing cap, if necessary. Pry the damaged pin out of the bearing cap. Use compressed air to blow any dirt or material out of the pin hole.

The pin insertion tool can be fabricated locally. The tool is to be made from 4340 steel, or equivalent. The material will need to be thoroughly hardened to 45-50 HRC. Fabricate the pin insertion tool using the following dimensions. The overall length of the tool is 160 mm [6.299 in].

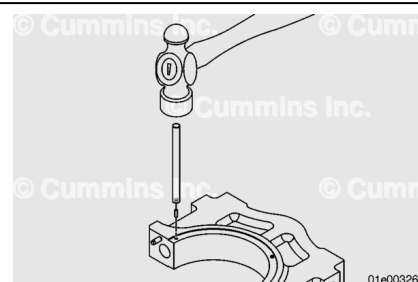
- A. 12.5 mm [0.492 in]
- B. 0.5 mm x 0.5mm [0.020 in x 0.020 in]
- C. 4.5 mm diameter [0.177 in]
- D. 3.2 mm deep [0.128 in].

Chamfer (1) should be no greater than 0.4 x 0.2 mm [0.016 x 0.008 in].

All dimensions +/- 0.2mm [0.008 in].

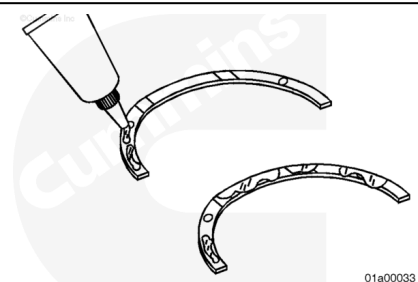


Install the new thrust bearing retaining pins (two on each side) into the main bearing cap to facilitate the installation of the lower thrust bearings. To avoid thrust bearing retaining pin damage, use the pin insertion tool and a copper hammer. This tool will set the pins to the correct depth of protrusion. Check the thrust bearing retaining pin protrusion after installation.



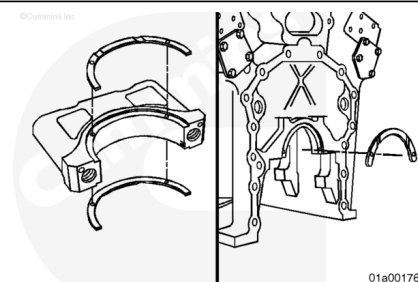
Install

Use engine oil. Lubricate the upper thrust bearings.



⚠ CAUTION ⚠

The grooves in the thrust bearings must point toward the crankshaft. The dowels that secure the bearings must not protrude above the bearing. If the grooves are toward the cylinder block or the dowels extend above the surface of the bearing, the crankshaft can be damaged beyond reuse.



Install the thrust bearings.

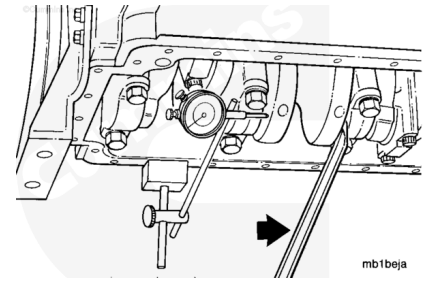
Install two thrust bearings on the number 7 main bearing cap and one on the number 7 saddle on the rear of the block as shown.

Use a dial indicator to measure the crankshaft end clearance.

Refer to Procedure 001-016 in Section 1.

(/qs3/pubsys2/xml/en/procedures/57/57-001-016-tr.html)

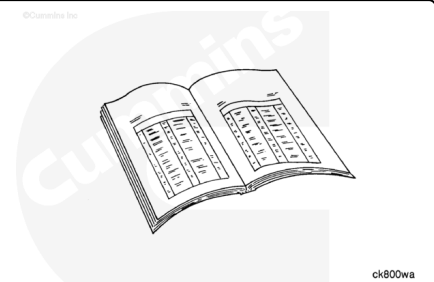
If the clearance is **not** within specifications, check for foreign material. Oversize thrust bearings are available to adjust the end clearance. Repair limit is 0.50 mm [0.0197 in].



Finishing Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



- Install the oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/57/57-007-035-tr.html)
- Install the oil pan and gasket. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/57/57-007-025-tr.html)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/57/57-007-037-tr.html)
- Operate the engine and check for leaks.

